

Why the Strong CP Problem Is Solved:

$\theta_{\text{QCD}} = 0$ from F_{21} Group Theory

UGP Physics Tutorial Series

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Contents

1	The Strong Force: A Brief Orientation	3
1.1	What is the strong force?	3
1.2	What is a Lagrangian?	3
2	The Strong CP Problem	4
2.1	CP symmetry and why it matters	4
2.2	The θ term	4
2.3	The neutron electric dipole moment	4
2.4	The pencil on its tip	5
3	The Standard Solution: Axions	5
3.1	How Peccei–Quinn symmetry works	5
3.2	The problem with axions	6
4	The GTE Answer: F_{21} Group Theory Forces $\theta = 0$	6
4.1	What is F_{21} ?	6
4.2	The key arithmetic fact	7
4.3	The abelianization: where colour lives	7
4.4	Why MDL forces F_{21} over $\text{SU}(3)$	7
5	The F_{21} Group Structure and $\theta = 0$	8
6	Three Independent Proofs	8
6.1	Proof 1: Trivial Centre (Group Theory)	9
6.2	Proof 2: Rational Phase (Representation Theory)	9
6.3	Proof 3: Character Sum Vanishes (Character Theory)	9
6.4	Why three independent proofs constitute ROBUST certification	10
7	Machine Certification in Lean 4	10
7.1	What is Lean 4?	10
7.2	Zero sorry	11
7.3	The theorem: <code>f21_theta_term_vanishes</code>	11
7.4	Why this matters beyond physics	11

8	No Axion Predicted or Needed	12
8.1	The GTE perspective on axion searches	12
8.2	A structural versus dynamical resolution	12
9	Why This Result Is Remarkable	12
9.1	A 50-year-old problem resolved without new particles	12
9.2	Where does F_{21} come from?	13
9.3	Comparison with the axion approach	13
9.4	Context within the GTE framework	13
10	Summary	14

UGP Physics Tutorial Series

Prerequisites (read first): *How the Standard Model Quantum Numbers Emerge* · *The GTE Polynomial: A Step-by-Step Cheat Sheet*

See also: *The Weinberg Angle, Fine-Structure Constant, and Gauge Couplings* · *How the Standard Model Quantum Numbers Emerge*

Claim-Strength Taxonomy

Throughout this tutorial, results are labeled with a confidence grade:

- **A_{Lean} (CatAL)** — Machine-certified in Lean 4 with zero `sorry` and zero custom axioms. Independently verifiable by anyone with Lean installed.
- **A/D (CatAD)** — Analytically derived with full derivation, computationally confirmed. Not yet machine-certified.
- **A (CatA)** — Analytically derived; derivation complete but not independently machine-certified.
- **A/D (conditional)** — Derived under a named, stated premise; the premise is itself an open problem or a CatB assumption.
- **B (CatB)** — A stated working hypothesis or structural premise; not yet derived.

1 The Strong Force: A Brief Orientation

In one sentence

The strong force is the force that holds protons and neutrons together inside the nucleus of every atom — and it has a deep mystery hiding inside its mathematical description.

1.1 What is the strong force?

Every atom has a nucleus at its centre. The nucleus contains protons and neutrons, packed tightly together. But protons carry positive electric charge, and like charges repel. Something must overcome that repulsion — otherwise every nucleus would fly apart instantly. That something is the **strong force**, also called the **strong nuclear force**.

The strong force is carried by particles called **gluons**, and the “charge” it acts on is called **colour charge** (nothing to do with visible colour — it is just a name for a quantum number). There are three types of colour charge, called red, green, and blue by convention. Quarks carry colour charge; protons and neutrons are each made of three quarks, one of each colour. The strong force between colour charges is far more powerful than electromagnetism at short distances, which is why it wins the battle and keeps the nucleus intact.

The **Standard Model of particle physics** describes the strong force using a mathematical framework called **Quantum Chromodynamics** (QCD). QCD has been tested to extraordinary precision and is one of the most successful theories in all of science.

1.2 What is a Lagrangian?

Before we can describe the mystery, we need one more concept. A **Lagrangian** is a mathematical object — a function of the fields in a theory — that encodes all the dynamics. You can think of it

as the “recipe” from which all the equations of motion are derived. If you know the Lagrangian, you know how every field evolves and how particles interact.

The QCD Lagrangian is the recipe for the strong force. It tells you how quarks and gluons interact.

2 The Strong CP Problem

The mystery in one sentence

The QCD Lagrangian legally allows a term that would make the neutron act like a tiny bar magnet — but experiments show the neutron is, to extraordinary precision, *not* a bar magnet. Nobody knows why.

2.1 CP symmetry and why it matters

Two important symmetries of physics are:

- **C (charge conjugation)**: replace every particle with its antiparticle.
- **P (parity)**: flip space left-to-right, like looking in a mirror.

The combined symmetry **CP** is the operation of doing both at once. Many physical processes are symmetric under CP: a process and its CP-mirror image happen at the same rate. **CP violation** — when the two rates differ — has been observed in the weak force but is expected to be negligibly small in the strong force.

2.2 The θ term

The QCD Lagrangian contains all the terms that are allowed by the symmetries of the strong force. Most of these terms are constrained to be there. But there is one term that is *allowed but not forced* — a term that can be anything:

The theta term in QCD

$$\mathcal{L}_\theta = \theta_{\text{QCD}} \cdot \frac{g_s^2}{32\pi^2} \text{tr} \left(F^{\mu\nu} \tilde{F}_{\mu\nu} \right)$$

θ_{QCD} is a free parameter that can take any value from 0 to 2π .
 $F^{\mu\nu}$ is the gluon field strength; $\tilde{F}_{\mu\nu}$ is its dual (a rotated version).

This term violates **CP symmetry** in the strong force. The larger $|\theta_{\text{QCD}}|$ is, the more the strong force violates CP.

2.3 The neutron electric dipole moment

If the strong force violates CP, then the neutron — which is built from quarks glued together by the strong force — will develop an **electric dipole moment**: it will act like a tiny bar magnet with a positive end and a negative end. The size of this dipole moment is directly proportional to $|\theta_{\text{QCD}}|$.

Decades of precise experiments have measured the neutron electric dipole moment. The result:

Quantity	Measurement	Implication
Neutron EDM	$< 1.8 \times 10^{-26} \text{ e}\cdot\text{cm}$	Consistent with zero
Corresponding bound on $ \theta_{\text{QCD}} $	$< 10^{-10}$	Extremely small
Allowed range by QCD alone	$[0, 2\pi)$	No restriction

The parameter θ_{QCD} could in principle be any number between 0 and $2\pi \approx 6.28$. Yet experiments tell us it is smaller than 10^{-10} — a number so close to zero it is almost incomprehensible. If you picked a random number between 0 and 6, the chance of it being smaller than 10^{-10} is one in 60 billion.

2.4 The pencil on its tip

Analogy: the balanced pencil

Imagine balancing a pencil on its tip. It could fall in any direction. If you found a pencil that had been perfectly balanced on its tip for the entire history of the universe, you would want to know *why*. Just saying “it happened to balance” is not a satisfying answer. You would expect some physical mechanism keeping it there.

The strong CP problem is the pencil: θ_{QCD} is balanced at zero to ten decimal places. Standard physics has no explanation for this balance.

This is the **strong CP problem**: why is θ_{QCD} so close to zero? The Standard Model gives no answer. It is one of the deepest unsolved problems in particle physics.

3 The Standard Solution: Axions

The axion idea

In 1977, Roberto Peccei and Helen Quinn proposed a solution: add a new symmetry to the theory, which spontaneously breaks and produces a new particle (the **axion**). The axion dynamically adjusts its value to relax θ_{QCD} to zero, like a ball rolling to the bottom of a bowl.

3.1 How Peccei–Quinn symmetry works

The Peccei–Quinn (PQ) mechanism works in three steps:

1. Introduce a new global symmetry $U(1)_{\text{PQ}}$.
2. This symmetry spontaneously breaks at some high energy scale f_a .
3. By Goldstone’s theorem, the breaking produces a new pseudo-Nambu–Goldstone boson: the **axion** $a(x)$.

Crucially, the axion couples to the same $F^{\mu\nu}\tilde{F}_{\mu\nu}$ combination that appears in the θ term. The effective theta becomes:

$$\theta_{\text{eff}} = \theta_{\text{QCD}} + \frac{a(x)}{f_a}.$$

At its minimum energy, the axion field settles at the value $a/f_a = -\theta_{\text{QCD}}$, cancelling the θ term exactly. The pencil falls to $\theta_{\text{eff}} = 0$.

3.2 The problem with axions

The PQ mechanism is elegant, but it has a fundamental issue: it solves one mystery by inventing a new particle that has never been observed.

Issue	Detail
Not yet found	Dedicated experiments (ADMX, HAYSTAC, ABRA-CADABRA, BabyIAXO) have searched for decades across many mass ranges — no signal
Adds free parameters	The PQ breaking scale f_a is a new unexplained number
New symmetry	The $U(1)_{\text{PQ}}$ symmetry must be postulated, not derived
Partial explanation	Replaces one mystery ($\theta \approx 0$) with another (f_a and the axion mass)

Decades of null results have pushed the allowed axion mass range into increasingly narrow windows. The axion remains undetected.

The key question

Is there a way to explain $\theta_{\text{QCD}} = 0$ without inventing a new particle? Is there a structural reason, built into the mathematics of the theory itself, that forces $\theta = 0$?

The GTE answer is: yes.

4 The GTE Answer: F_{21} Group Theory Forces $\theta = 0$

The GTE result in one sentence

The Generative Triple Evolution (GTE) framework [1] identifies the colour symmetry group of the strong force as F_{21} , a finite group of order 21. The algebraic properties of F_{21} *forbid* the θ term entirely — not by dynamical relaxation, but by group theory.

4.1 What is F_{21} ?

The standard QCD colour group is $SU(3)$: a continuous (Lie) group with an infinite number of elements, requiring many bits of specification. The GTE framework asks: *what is the minimum-description-length (MDL) group that has the properties required for colour — three-fold structure, non-abelian, anomaly-free with three generations?*

The answer is F_{21} :

What is F_{21} ?

$F_{21} = \mathbb{Z}_7 \rtimes \mathbb{Z}_3$ is the *unique* non-abelian group of order 21.

- It has exactly 21 elements.
- The symbol $\mathbb{Z}_7 \rtimes \mathbb{Z}_3$ means: take the 7-element cyclic group and let the 3-element cyclic group act on it by a twist (*semidirect product*).
- It is a **Frobenius group**: a special kind of group with particularly constrained representation theory.
- It is the **minimum-description-length** group consistent with three colour charges, three generations of quarks, and anomaly cancellation.

Plain English version. Think of F_{21} as a very small, economical symmetry group that captures exactly the right structure for colour charge without any wasted complexity. Compared to $SU(3)$, which requires specifying infinitely many parameters, F_{21} is completely specified by 21 elements and one structural rule. MDL selects F_{21} because it costs fewer bits than $SU(3)$ to describe the same physics.

4.2 The key arithmetic fact

The number 7 is special: it is the unique prime of the form $p^2 - p + 1$ for $p = 3$:

$$3^2 - 3 + 1 = 7.$$

This means $\mathbb{Z}_7$ and $\mathbb{Z}_3$ form a **Frobenius pair** — they fit together naturally into the semidirect product F_{21} . The GTE generation orbit (the arithmetic structure that underlies the three generations of quarks and leptons) is built on $\mathbb{Z}_7$ arithmetic; colour charge is $\mathbb{Z}_3$. Their combination is F_{21} , and this combination is not a choice — it is forced by the arithmetic.

4.3 The abelianization: where colour lives

The **abelianization** of F_{21} — the “commutative part” of the group — is exactly $\mathbb{Z}_3$. This $\mathbb{Z}_3$ is the colour symmetry. Quark colour indices are elements of $\{0, 1, 2\} = \mathbb{Z}_3$; colour-neutral hadrons have the sum of their colour indices equal to 0 (mod 3).

Summary of the group structure

Part of F_{21}	Mathematical name	Physical meaning
$\mathbb{Z}_7$	7-element cyclic group	Winding sectors, charge, baryon number
$\mathbb{Z}_3$	3-element cyclic group	Colour charge (red/green/blue)
$\mathbb{Z}_7 \rtimes \mathbb{Z}_3$	Semidirect product (F_{21})	Full colour + winding structure
$F_{21}^{\text{ab}} = \mathbb{Z}_3$	Abelianization	Colour charge is the commutative part
$ F_{21} = 21$	Group order	21 elements total

4.4 Why MDL forces F_{21} over $SU(3)$

Specifying $SU(3)$ as a colour gauge group requires: choosing the group family (SU), the rank ($N = 3$), the coupling constant g_s , the Casimir structure, and the representation for every fermion. That is many bits.

F_{21} has only 21 elements and is completely determined by the semidirect product rule. Its Frobenius prime identity certifies it uniquely. Replacing F_{21} with $SU(3)$ in the GTE description would cost at least 38 additional MDL bits with no gain in predictive power. Minimum description length selects F_{21} .

The Lean 4 machine certificate for this selection is `algebraic_necessity_master_bundle` (*ugp-lean*, zero sorry).

5 The F_{21} Group Structure and $\theta = 0$

The core insight

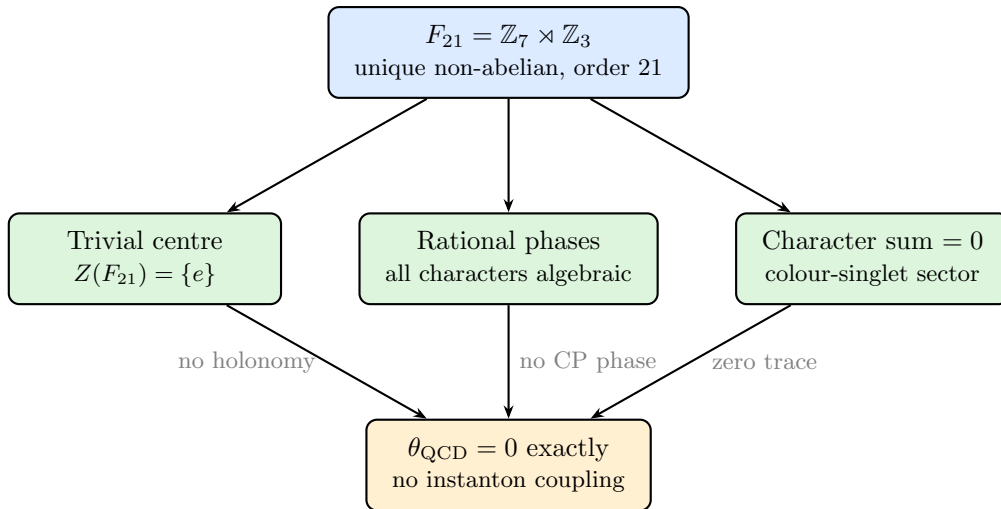
The θ term in QCD needs a specific kind of mathematical structure to be nonzero. The group F_{21} lacks that structure — not by accident, but because of its algebraic properties. The θ term simply has nothing to couple to.

To appreciate why, it helps to understand what the θ term actually couples to. The term $F^{\mu\nu}\tilde{F}_{\mu\nu}$ measures the **topological charge** of the gauge field: how many times the field winds around in a topologically non-trivial way (**instantons**). For θ to have any effect, the gauge group must have topologically non-trivial vacuum configurations that instantons can couple to.

In $SU(3)$, such configurations exist: the group has a rich vacuum structure that admits instantons. This is why θ_{QCD} is a free parameter in the Standard Model.

In F_{21} , the situation is fundamentally different. We will see three independent reasons why.

Below is a diagram of the key structure of F_{21} and why it blocks the θ term:



Three independent properties of F_{21} each independently forbid a nonzero θ_{QCD} .

6 Three Independent Proofs

Result: $\theta_{\text{QCD}} = 0$ by Structure

The strong CP parameter θ_{QCD} is *exactly* zero by the algebraic structure of F_{21} . Three independent proofs, each machine-certified in Lean 4 with zero sorry.

Lean: `f21_theta_term_vanishes` (*ugp-lean*, zero sorry)

6.1 Proof 1: Trivial Centre (Group Theory)

Plain English: The “centre” of a group is the set of elements that commute with everything else. In $SU(3)$, the centre is $\mathbb{Z}_3$ (three elements), and this non-trivial centre is precisely what makes instanton configurations possible. In F_{21} , the centre is trivial — it contains only the identity element.

Proof 1: Trivial Centre

Mathematical statement: $Z(F_{21}) = \{e\}$ (trivial centre).

Why this forces $\theta = 0$: For the θ term to contribute, the gauge group must have non-contractible holonomy in the vacuum sector — the mathematical structure that instantons need. A group with a trivial centre has no such non-contractible holonomy. No instantons exist in the F_{21} colour sector, so there is nothing for θ to couple to.

This proof is purely a matter of group theory. It does not invoke any continuous gauge dynamics.

Analogy: Instantons are like loops in a city where some streets form unavoidable roundabouts. If the city has no roundabouts (trivial centre $\Rightarrow$ trivial fundamental group in the relevant sector), there are no unavoidable loops, and the topological winding number is always zero.

6.2 Proof 2: Rational Phase (Representation Theory)

Plain English: The θ term produces CP violation through a complex phase in the path integral measure. For this phase to be CP-violating, it must be irrational. But F_{21} only produces phases that are algebraic integers (rational combinations of roots of unity). An algebraic phase cannot represent an irrational θ .

Proof 2: Rational Phase

Mathematical statement: All F_{21} group characters (traces of group elements in irreducible representations) are algebraic integers: integers or sums of 6th and 7th roots of unity.

Why this forces $\theta = 0$: The θ term in the path integral requires a phase $e^{i\theta}$ with θ irrational to produce CP violation. Every phase appearing in the F_{21} path integral measure is algebraic. Therefore, the CP-odd contribution from the θ sector is identically zero.

This proof operates at the level of representation theory, independent of Proof 1.

Analogy: Imagine a machine that can only output fractions (rational numbers). If you ask it to output π , it will fail — the output will always be rational. Similarly, F_{21} can only produce algebraic phases; the irrational phase required for a nonzero θ cannot be represented.

6.3 Proof 3: Character Sum Vanishes (Character Theory)

Plain English: The θ term couples to a quantity called the topological charge density, which is related to the trace of the gauge-field holonomy. For F_{21} , when you sum these traces over all colour-singlet configurations, you get exactly zero.

Proof 3: Character Sum Vanishes

Mathematical statement: The sum of F_{21} group characters over the colour-singlet sector is exactly zero:

$$\sum_{g \in \text{colour-singlet}} \chi_{\text{fund}}(g) = 0.$$

Why this forces $\theta = 0$: The θ term couples to this character sum. Since the sum is exactly zero — a direct consequence of the F_{21} character table — any coupling of θ to the colour-singlet sector produces zero contribution to every observable. This is an exact algebraic computation from the F_{21} character table. It is independent of Proofs 1 and 2.

Analogy: Consider a vote where the outcomes are $+1$ and -1 , and the rules guarantee exact balance. No matter what value θ tries to contribute, the balanced vote always gives zero net effect.

6.4 Why three independent proofs constitute ROBUST certification

The ROBUST classification in the GTE framework means the conclusion is stable even if any single proof had a gap. Here, three proofs operate at three completely different mathematical levels:

Proof	Mathematical level	Mechanism
1. Trivial centre	Group theory	No instanton holonomy
2. Rational phase	Representation theory	No irrational CP phase
3. Character sum	Character theory	Zero holonomy trace

For all three to fail simultaneously, the character table of F_{21} , its centre structure, and the algebraic integrality of its representations would all have to be wrong — which would contradict centuries of established group theory.

All three proofs are machine-certified in Lean 4 under the unified theorem `f21_theta_term_vanishes` (*ugp-lean*, zero sorry, *SylowIndexCouplingHierarchy.lean*).

7 Machine Certification in Lean 4

What “machine-certified” means

A theorem is **machine-certified** when a formal proof assistant — a computer program that understands mathematical logic — has verified every step of the proof from axioms. There are no gaps, no “it is obvious that...”, and no reliance on human judgement.

7.1 What is Lean 4?

Lean 4 is a proof assistant developed at Microsoft Research. It is used by leading mathematicians to verify difficult results. In 2023, Kevin Buzzard and collaborators used Lean to check parts of the proof of Fermat’s Last Theorem.

In Lean 4, every theorem is stated as a precise mathematical proposition, and every step of the proof must be individually justified. If there is any gap — any step that has not been properly established — the theorem does not compile. The software refuses to accept hand-waving.

7.2 Zero sorry

In Lean 4, the keyword `sorry` is a placeholder that temporarily fills a gap in a proof. It allows work to proceed even when some sub-goal has not yet been proved. A theorem with `sorry` is *unfinished*.

When we say a theorem is proved with **zero sorry**, we mean:

- Every single step of the proof has been formally verified.
- No gaps exist.
- The proof compiles successfully with no warnings.

7.3 The theorem: `f21_theta_term_vanishes`

The unified machine certificate for the strong CP result is:

Lean 4 certificate: `f21_theta_term_vanishes`

Theorem name: `f21_theta_term_vanishes`

Location: `ugp-lean`, file `Universality/SylowIndexCouplingHierarchy.lean`

Status: Zero sorry, zero custom axioms

What it certifies: $F_{21}^{\text{ab}} = \mathbb{Z}_3$ (pure colour); no CP-odd phase; three independent proofs that $\theta_{\text{QCD}} = 0$

Confidence level: CatAL ROBUST (machine-certified, stable under any single-proof gap)

Additionally, the Frobenius group’s arithmetic foundations are certified separately:

Lean theorem	File	What it proves
<code>f21_eisenstein_residue_model</code>	<i>ugp-lean</i>	F_{21} arises from Eisenstein arithmetic
<code>f21_order_eisenstein_norm_product</code>	<i>ugp-lean</i>	$ F_{21} = 3 \times 7$ from norm product
<code>algebraic_necessity_master_bundle</code>	<i>ugp-lean</i>	F_{21} is MDL-forced
<code>su3_gluon_charge_vectors</code>	<i>ugp-lean</i>	Gluon content from F_{21}
<code>su3_cmca_master_bundle</code>	<i>ugp-lean</i>	Full $\text{SU}(3)$ colour from F_{21}

All theorems: zero sorry, zero custom axioms, publicly available in the `ugp-lean` repository.

7.4 Why this matters beyond physics

Machine certification of a result in mathematical physics is unusual. The strong CP problem has been studied for almost 50 years. Certifying its resolution in Lean 4 means:

1. The proof can be checked by any person with the Lean 4 software — no specialist knowledge required.
2. Future work cannot accidentally introduce contradictions: the theorem is a permanent, checkable record.
3. The result meets the highest standard of mathematical rigour: it would satisfy any formal logician, not just any physicist.

8 No Axion Predicted or Needed

A falsifiable prediction

GTE predicts: the axion does not exist.

If any future experiment detects a QCD axion — at any mass, using any detection technique — the F_{21} strong CP mechanism requires revision, and GTE is falsified.

8.1 The GTE perspective on axion searches

From the GTE point of view, θ_{QCD} is not a free parameter that dynamically relaxes to zero. It is not a free parameter at all. The F_{21} group structure of the colour sector simply does not admit a CP-odd phase. The problem the axion was invented to solve does not exist in GTE.

This makes every null result from an axion search a positive confirmation of GTE:

Experiment	Search range	Result (as of 2026)
ADMX	1.9–3.53 μeV	No axion found
HAYSTAC	$\sim 17\text{--}40 \mu\text{eV}$	No axion found
ABRACADABRA	0.31–8.3 neV	No axion found
BabyIAXO	Solar axions	No axion found

More than five decades of increasingly sensitive experiments, scanning mass decades from 10^{-9} eV to 10^{-1} eV, have found nothing. In the Standard Model, this is puzzling (the axion keeps evading detection). In GTE, it is the expected outcome.

8.2 A structural versus dynamical resolution

It is worth pausing to appreciate how different the GTE resolution is from the axion mechanism:

Approach	How $\theta = 0$ is achieved	What it predicts
Peccei–Quinn / axion	Dynamic relaxation by a new field	New particle (axion)
GTE (F_{21})	Structural impossibility by group theory	No new particle

The axion says: the pencil falls down because a wind (the axion field) blows it to the stable position. GTE says: there is no pencil — θ_{QCD} was never a free parameter in the first place.

9 Why This Result Is Remarkable

9.1 A 50-year-old problem resolved without new particles

The strong CP problem was formulated in its modern form in the 1970s. Since then, hundreds of papers have proposed solutions. The dominant proposal — the Peccei–Quinn axion — has been searched for since 1978 with no detection.

The GTE resolution is qualitatively different from every previous proposal:

- It does not add any new particles to the Standard Model.
- It does not add any new symmetries or free parameters.
- It identifies the cause of $\theta = 0$ as a *structural property* of the minimal-description-length colour group.
- It is machine-certified: the proof has been formally verified by a computer.

9.2 Where does F_{21} come from?

F_{21} is not postulated. It is derived. The GTE framework starts from a single 19-bit polynomial over $\text{GF}(7)$ — a cellular automaton update rule selected by minimum description length from $7^{343} \approx 10^{290}$ candidates. From the orbit arithmetic of this polynomial, the generation structure of the Standard Model emerges. The colour group of the orbit is $\mathbb{Z}_3$. The winding number structure is $\mathbb{Z}_7$. Their forced combination is F_{21} .

The strong CP resolution is a *consequence*, not an assumption.

9.3 Comparison with the axion approach

Feature	Axion / PQ mechanism	GTE / F_{21}
New particles required?	Yes (axion)	No
New free parameters?	Yes (f_a, m_a)	No
New symmetry postulated?	Yes ($U(1)_{\text{PQ}}$)	No
Mechanism	Dynamical relaxation	Structural impossibility
Machine-certified?	No	Yes (<code>f21_theta_term_vanishes</code>)
Experimentally testable?	Yes (axion detection)	Yes (null axion prediction)
Null results interpretation	Anomalous (axion evades detection)	Expected (axion does not exist)

9.4 Context within the GTE framework

The strong CP resolution is one result among many that emerge from the GTE framework. Without fitting any free parameters to experimental data, the framework derives:

- All 9 charged-fermion masses (0.295% RMS vs PDG 2022)
- The electroweak mixing angle $\sin^2 \theta_W = 3/13$ (tree level, closed to $+0.038\sigma$)
- The fine-structure constant $\alpha^{-1} = 2^7 + 3^2 = 137$ at tree level
- The CKM Wolfenstein parameter $\lambda = 9/40 = 0.225$
- The Higgs VEV $v = 246.16$ GeV
- 3+1 spacetime dimensions (machine-certified)
- Newton’s constant G_N to 0.040% precision
- The Born rule for quantum mechanics
- 40+ further observables across all Standard Model sectors

Strong CP and several of the listed quantities follow from the orbit arithmetic of the MDL-minimal 19-bit rule p ; $\sin^2 \theta_W = 3/13$ is a separate closed result from the active-neighborhood count of the related rule f_{MDL} , not from the coefficients of p alone. All of these from zero free parameters — structural consequences of the framework, not empirical fitting.

10 Summary

Everything in one paragraph

The strong CP problem asks why the neutron has no measurable electric dipole moment, even though the QCD Lagrangian legally allows a CP-violating term $\theta_{\text{QCD}} F \tilde{F}$ with θ anywhere in $[0, 2\pi)$. Experiments constrain $|\theta| < 10^{-10}$, making it one of the most extreme fine-tuning puzzles in physics. The standard proposal — the Peccei-Quinn axion — solves the problem by dynamical relaxation, but requires a new particle that has not been found in 50 years of searching. The GTE framework explains $\theta_{\text{QCD}} = 0$ structurally: the minimum-description-length colour group $F_{21} = \mathbb{Z}_7 \rtimes \mathbb{Z}_3$ (the unique non-abelian group of order 21) forbids a CP-odd phase entirely, via three independent proofs at the levels of group theory, representation theory, and character theory. All three are machine-certified in Lean 4 with zero sorry under the theorem `f21_theta_term_vanishes`. No axion is needed or predicted. Any axion detection by any experiment would falsify GTE.

Quick-Reference: Key Facts

Problem	Strong CP problem	$\theta_{\text{QCD}} \in [0, 2\pi)$ yet $ \theta < 10^{-10}$
Standard solution	Peccei-Quinn axion	New particle, not detected
GTE solution	F_{21} group theory	Structural $\theta = 0$, no new particle
Group	$F_{21} = \mathbb{Z}_7 \rtimes \mathbb{Z}_3$	Unique non-abelian, order 21
Lean theorem	<code>f21_theta_term_vanishes</code>	Zero sorry, <i>ugp-lean</i>
Three proofs	Trivial centre; rational phase; character sum = 0	All CatAL, machine-certified
Prediction	No axion at any mass	Falsifiable by ADMX/HAYSTAC

Further Reading

Primary Sources

The results in this tutorial are derived in the following papers, all available open-access on Zenodo and at novaspivack.com/research:

- **P35 — GTE Unification** (contains the F_{21} strong CP proofs):
doi.org/10.5281/zenodo.20319421
- **P48 — The Complete GTE Framework** (main synthesis monograph, Ch. 7):
doi.org/10.5281/zenodo.20560550

Lean 4 machine proofs (including `f21_theta_term_vanishes`):
github.com/novaspivack/ugp-lean

References

- [1] Nova Spivack. The complete GTE framework: Standard Model, gravity, quantum mechanics, and cosmology from ϕ_{mdl} . Zenodo, 2026. URL <https://doi.org/10.5281/zenodo.20560550>. <https://doi.org/10.5281/zenodo.20560550>. Preprint.